PADHAKU
ICT – Class 6 · Class 6 ICT



CHAPTER 1

Introduction to ICT Environment

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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⭐ Practice and Activities

Practice 1 **Find the different hardware and software in your computer lab and paste their pictures in your notebook.**

💡 SAMPLE ANSWER

Hardware (things we can see and touch): monitor, keyboard, mouse, CPU, printer, speakers, pen drive, CD, hard disk.

Software (programs that make the devices work): Windows / Linux (operating system), a word processor, a presentation program, Paint, GCompris.

Cut or print their pictures and paste them under two headings in your notebook: Hardware and Software.

Practice 2 **Find pictures of examples of the primary storage unit and the secondary storage unit and paste them in your notebook.**

💡 SAMPLE ANSWER

Primary storage unit: RAM chip / RAM stick, ROM chip.

Secondary storage unit: hard disk, CD, DVD, pen drive.

Paste the pictures under these two headings.

Activity 1 **In GCompris, play the 'Complete the puzzle' game: drag the objects with the mouse and match them to their shapes.**

💡 SAMPLE ANSWER

Open GCompris and choose 'Complete the puzzle'. Click '?' to read the instructions.

Press the left mouse button on an object, keep it pressed, drag the object onto the shape that matches it, and release the button.

Match all objects one by one. This builds mouse skills (click and drag).

Practice 3 **Learn to use the keyboard and mouse by playing the 'Baby keyboard', 'Baby Mouse' and 'A word processor' activities in GCompris.**

💡 SAMPLE ANSWER

Baby keyboard: press any key and see the letter or number appear on the screen.

Baby Mouse: move the mouse and click on the screen as the game asks.

A word processor: type a few words or sentences, using Space, Enter and Backspace.

Follow the instructions given in each game.

Activity 2 **Open the 'Maze' game in GCompris and find the correct path through the maze.**

💡 **SAMPLE ANSWER**

Open the Maze game and click '?' for instructions.

Move the character with the arrow keys (or the mouse) and find the path from the start to the exit.

If a path is blocked, go back and try another way.

Activity 3 **Learn to play Sudoku in GCompris.**

💡 **SAMPLE ANSWER**

Open Sudoku and click '?' to read the rules.

Fill the grid so that each symbol or number comes only once in every row, every column and every box.

Start with the rows or columns that have the fewest empty places.

Practice 4 **Solve the 'Family Tree' puzzle using logic.**

💡 **SAMPLE ANSWER**

Open the Family Tree game and click '?' for instructions.

Look at how the people in the tree are joined (parents, children, husband–wife).

Use this to choose the correct relation (e.g. father, sister, grandmother) for the question asked.

Practice 5 **Make a poster with the do's and don'ts and put it up in a prominent place in your classroom or computer room.**

💡 SAMPLE ANSWER

DO's

- Take the teacher's permission before using the computer.
- Follow all the steps to switch on the system.
- Use the keyboard and mouse gently; keep the mouse on the mouse pad.
- Tell the person in charge about any problem with the computer.
- Keep your back straight while working.
- Shut down the computer properly.

DON'Ts

- Don't bring food or drinks into the computer room.
- Don't touch any part of the computer with wet hands.
- Don't press the keys too hard.
- Don't damage, remove or disconnect any label, part, cable or device.
- Don't install or download software or change/delete system files.
- Don't make noise in the computer room.

Exercises · 1. Multiple choice questions

i **We can contact our friends by:**

a. e-mail b. chatting c. video conferencing d. all three

✔ ANSWER

d. all three

E-mail, chatting and video conferencing are all ways to contact friends.

ii **The keyboard is an example of:**

a. output unit b. input unit c. CPU d. none of these

✔ ANSWER

b. input unit

We use the keyboard to enter data and give commands to the computer.

iii **An example of an output unit is:**

a. printer b. monitor c. mouse d. both A and B

✔ ANSWER

d. both A and B

Processed data is shown on the monitor and printed by the printer. The mouse is an input device.

iv **What can be called the brain of the computer?**

a. speaker b. CPU c. monitor d. keyboard

✔ ANSWER

b. CPU

All the decisions of the computer are taken in the CPU.

v **Choose the names of famous operating systems:**

a. Windows b. Linux c. monitor d. both A and B

✔ ANSWER

d. both A and B

Windows and Linux are operating systems; a monitor is hardware.

📝 Exercises · 2. Fill in the blanks

a **ICT means _____.**

✔ ANSWER

ICT means **Information and Communication Technology**.

b **The Save option is in the _____ menu.**

✔ ANSWER

The Save option is in the **File** menu.

c **In the _____, information is stored temporarily.**

✔ ANSWER

In the **primary storage unit (RAM)**, information is stored temporarily.

d **The first program that starts in the computer when we press the Start button is called _____.**

✔ ANSWER

It is called the **Operating System**.

e **Smartphones and tablets are examples of _____.**

✔ ANSWER

Smartphones and tablets are examples of **palmtops**.

📝 Exercises · 3. Mark ✔ for true and × for false

a **A computer is an electronic machine.**

✔ ANSWER

✔ **True**

b **We can touch any part of the computer with wet hands.**

✔ ANSWER

× **False.** We must never touch any part of the computer with wet hands.

c **A laptop is bigger in size than a desktop.**

✔ ANSWER

× **False.** A laptop is smaller; we can keep it on our lap. A desktop is kept on a table.

d **ICT is not helpful in any area of our daily life.**

✔ ANSWER

× **False.** ICT helps in every area of daily life, such as education, health and banking.

e **We can see and touch a nibble.**

✔ ANSWER

× **False.** A nibble is a group of 4 bits, a unit of memory. It is not hardware, so we cannot see or touch it.

❓ Exercises · 4. Short answer questions

1 **Define a computer.**

✔ ANSWER

A computer is an electronic machine that stores information and helps us to manipulate (process) it. It can do many kinds of work according to the commands we give it.

2 **Name any four areas of our daily life where computers are used a lot.**

✔ ANSWER

- **Education:** learning in an effective and interesting way.
- **Hospitals:** finding diseases and treating patients.
- **Banking:** easy transfer of money.
- **Travel:** information and ticket booking for rail, air and road travel.

(Also: offices, e-mail and video calls, entertainment, science, art and design.)

3 **On the basis of size, how many types of computers can there be? Name them.**

✔ ANSWER

On the basis of size, computers are of **three** types:

- **Desktop:** used by keeping it on a table.

- **Laptop:** can be used by keeping it on our lap.
- **Palmtop:** can be used on the palm of the hand, e.g. tablets, smartphones.

4 **Write a short note on the operating system.**

✔ ANSWER

The operating system is the first program that starts when we press the Start button of the computer.

It is a link between the user and the computer: it takes commands from the user to the computer and brings the computer's answer back to the user.

It is a group of programs needed for the basic working of the computer, and it makes a suitable environment for other programs to run.

Examples: **Windows, Linux**.

5 **Briefly explain bytes.**

✔ ANSWER

A **byte** is a group of **8 bits**.

It is the smallest unit that can represent one data item or one character (letter, digit or symbol).

Storage capacity of memory is measured in bytes: 1 KB = 1024 bytes, 1 MB = 1024 KB, 1 GB = 1024 MB.

❓ Exercises · 5. Long answer questions

1 Explain the memory unit in detail.

✔ ANSWER

A **memory unit** is the amount of data that can be stored in a storage unit. Storage capacity is expressed in **bytes**.

Main memory units:

- **Bit (Binary Digit):** a binary digit, logical 0 or 1. It shows the off (inactive) or on (active) state of a part in an electric circuit.
- **Nibble:** a group of 4 bits.
- **Byte:** a group of 8 bits; the smallest unit that can represent one data item or one character.

Higher units:

Unit	Equal to
Kilobyte (KB)	1024 Bytes
Megabyte (MB)	1024 KB
Gigabyte (GB)	1024 MB
Terabyte (TB)	1024 GB
Petabyte (PB)	1024 TB

Example: a phone with 4 GB RAM has 4 × 1024 MB = 4096 MB of RAM.

2 (a) Distinguish between hardware and software.

✔ ANSWER

Hardware	Software
Physical parts that we can see and touch	A group of programs that make a device work
e.g. monitor, mouse, keyboard, hard disk, CD, pen drive, mobile phone	e.g. Windows, Linux, presentation software, painting software
Cannot do any work without software	Needs hardware to run

2 (b) **Distinguish between files and folders.**

✔ ANSWER

File	Folder
All the information or data about a topic stored at one place	Many files kept together at one place
Holds data, e.g. a text document named 'Introduction'	Helps keep files safe and well arranged, e.g. a folder named 'ICT'
Made with New → Text Document (or a program)	Made with New → Folder; it looks like a yellow folder icon

2 (c) **Distinguish between the primary storage unit and the secondary storage unit.**

✔ ANSWER

Primary storage unit	Secondary storage unit
Information is stored here temporarily first	Information is stored here permanently

Types: RAM (Random Access Memory) and ROM (Read Only Memory)	Examples: hard disk, CD, DVD, pen drive

2 (d) **Distinguish between a byte and a nibble.**

✔ ANSWER

Byte	Nibble
A group of 8 bits	A group of 4 bits
Smallest unit that can represent one character or data item	Half of a byte
1 byte = 2 nibbles	1 nibble = ½ byte

3 **Write the steps to rename a file.**

✔ ANSWER

Step 1: Bring the mouse pointer on the file and press the **right** mouse button. A menu opens.

Step 2: Choose the **Rename** option. The old name gets selected.

Step 3: Type the new name with the keyboard.

Step 4: Press **Enter** (or click anywhere outside). The file now has the new name.

(Shortcut: select the file and press **F2**.)

🎉 **Chapter 1 done! Revise the answers once more tomorrow.**